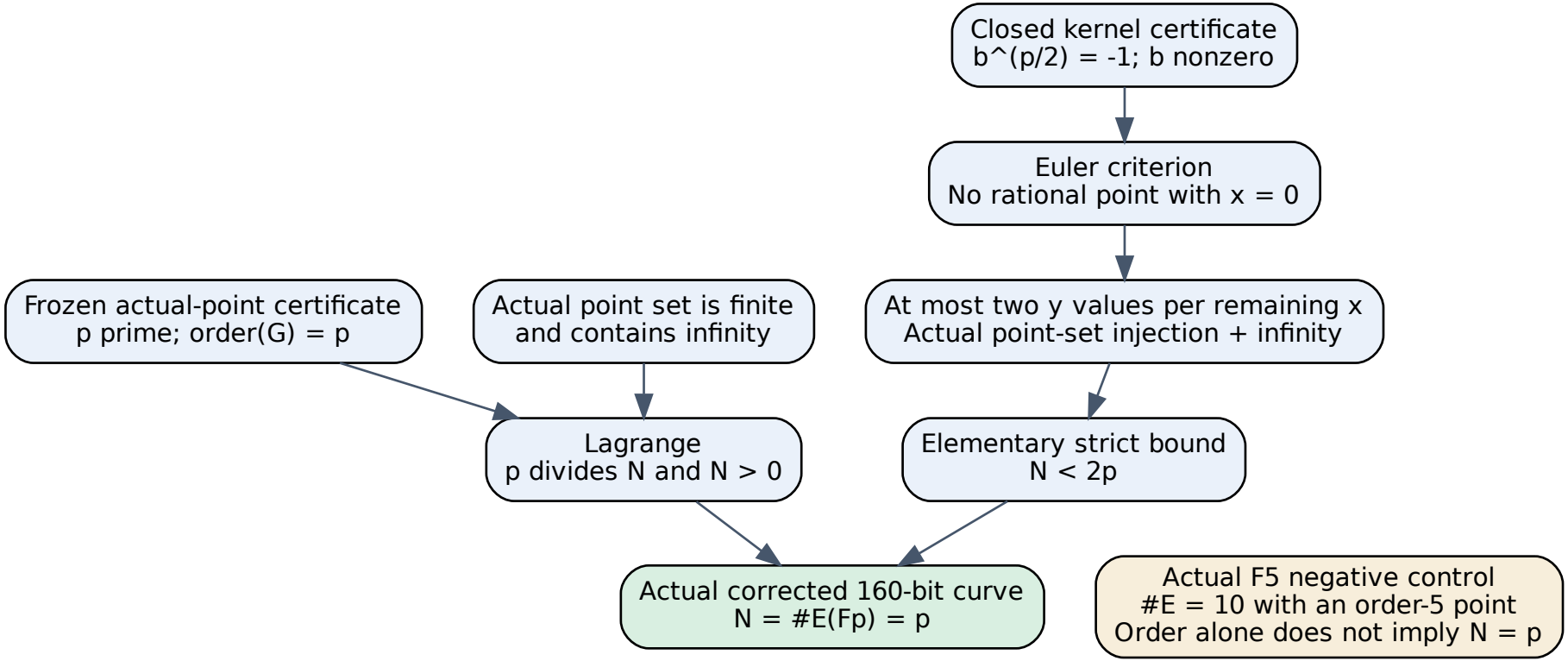
Closed kernel certificate
b^(p/2) = -1; b nonzero
Euler criterion
No rational point with x = 0
Frozen actual-point certificate
p prime; order(G) = p
Actual point set is finite
and contains infinity
At most two y values per remaining x
Actual point-set injection + infinity
Lagrange
p divides N and N > 0
Elementary strict bound
N < 2p
Actual corrected 160-bit curve
N = #E(Fp) = p
Actual F5 negative control
#E = 10 with an order-5 point
Order alone does not imply N = p